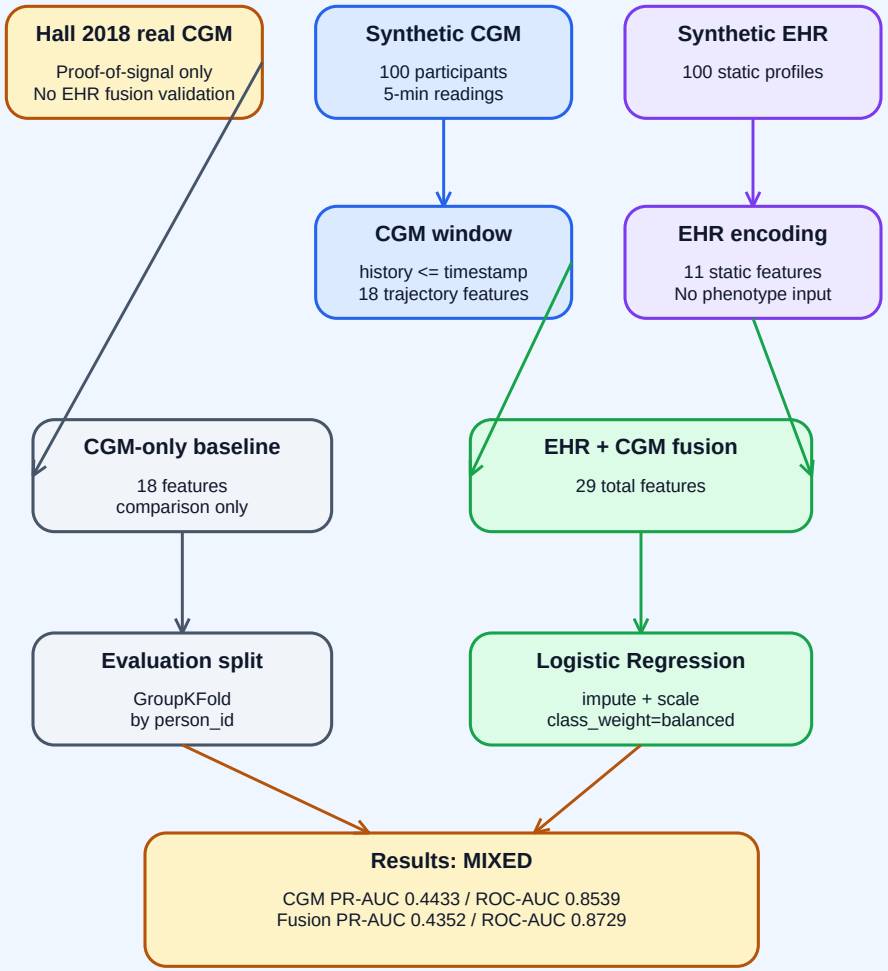
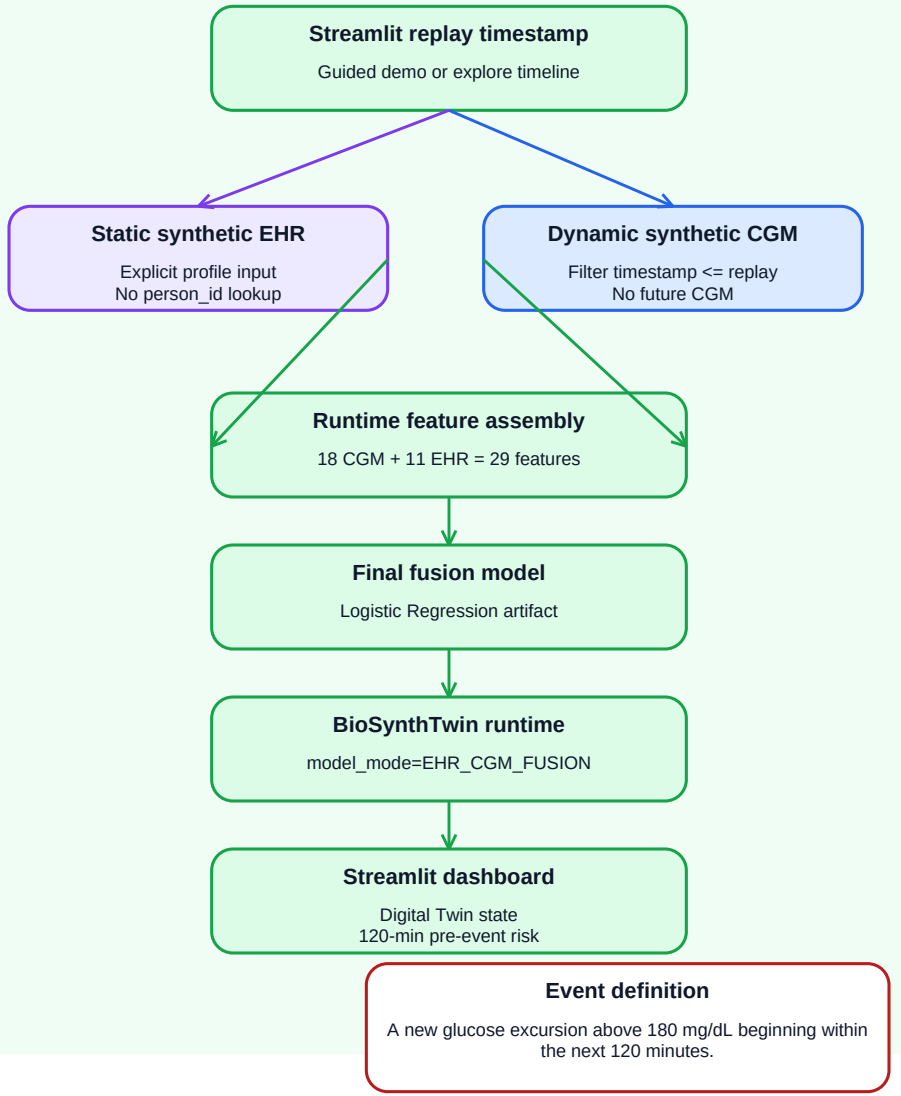
BioSynth Digital Twin Architecture

Offline training/evaluation is separated from live replay inference.

Offline training and evaluation



Live replay and inference



BioSynth Digital Twin Architecture

Data provenance, metrics, and runtime boundaries.

Offline pipeline

- Hall 2018 real CGM supports CGM trajectory proof-of-signal only.
- Synthetic CGM supplies 5-minute readings for 100 virtual participants.
- Synthetic EHR supplies one static profile for each synthetic participant.
- GroupKFold evaluation reports out-of-fold metrics.

Live replay pipeline

- Dashboard filters CGM to timestamp <= replay timestamp.
- Fusion mode requires an explicit static EHR profile.
- BioSynthTwin assembles features in saved metadata order.
- Streamlit displays Digital Twin state and 120-minute pre-event risk.

Key synthetic participant-separated metrics: MIXED

Model	Feature stream	PR-AUC	ROC-AUC
CGM trajectory	Dynamic CGM only	0.4433	0.8539
EHR + CGM fusion	Static EHR + dynamic CGM	0.4352	0.8729

Fusion improved ROC-AUC but slightly reduced overall PR-AUC compared with the CGM trajectory model.

Runtime exclusions

- No future CGM enters inference.
- No future event labels enter inference.
- person_id and simulation_phenotype are not model inputs.
- Simulator ground-truth parameters are not model inputs.

Limitations

- Synthetic EHR is phenotype-aligned and not a real clinical record.
- Synthetic CGM is simulator-generated.
- Hall 2018 did not validate EHR fusion.
- POC is not clinically validated or intended for medical decision-making.